

Structure-informed machine learning for drug discovery: a task-centric perspective

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Abstract

Advances in protein structure prediction have transformed the landscape of structure-based drug discovery, enabling deep learning models to incorporate spatial constraints into the design of target-specific ligands. This review provides a comprehensive synthesis of structure-aware molecular modeling from a task-centric perspective, focusing on binding pocket identification, interaction prediction, pose estimation, and complex modeling. We highlight the technological evolution from traditional docking and scoring frameworks toward geometry-informed deep learning architectures that encode protein structures via surface geometry, equivariant representations, and multi-modal embeddings. Special attention is given to recent progress in structure-conditioned molecular generation. We classify generative approaches into four core strategies: sequence-based generation with 3D conditioning, fragment-based linking and growing, graph-based generation under structural constraints, and 3D coordinate-based generation including diffusion models. Each paradigm balances chemical validity, spatial fidelity, and computational tractability in distinct ways, with diffusion-based and point cloud models emerging as powerful tools for synthesizing pocket-complementary molecules in full 3D space. We also discuss the emergence of co-folding models, which unify protein folding and ligand binding into a single predictive framework, bridging the gap between sequence-level learning and structural resolution. Finally, we examine the key challenges of data scarcity, generalization, and multi-objective control, and outline future directions toward scalable, interpretable, and physically plausible generation pipelines. By tracing how structural knowledge is reshaping AI-driven drug design, this review aims to provide both a conceptual roadmap and practical insight into next-generation molecular modeling.

Keywords drug discovery, machine learning, protein structure modeling, drug–target interaction

Introduction

Drug discovery has long been a complex, resource-intensive endeavor, characterized by low hit rates, high development costs, and substantial experimental uncertainties. Central to rational drug design is the structural understanding of protein targets, which enables the identification of novel druggable pockets (e.g. the allosteric pocket of Kirsten rat sarcoma viral oncogene homolog G12C) [1] and elucidation of resistance mutations (e.g. the T790M in epidermal growth factor receptor) [2]. However, for decades, progress in structure-based design was limited by the scarcity and cost of experimentally resolved

protein structures, typically obtained through X-ray crystallography or cryo-electron microscopy.

This landscape has been fundamentally reshaped by recent breakthroughs in protein structure prediction. Deep learning models such as AlphaFold2 [3], ESMFold [4], and RoseTTAFold [5] have transformed protein structures from rare experimental assets into accessible, proteome-wide computational resources. AlphaFold3 [6] further expands this capacity by supporting multimolecular complex modeling—including interactions with ligands, RNA, and DNA—enabling end-to-end prediction of drug–target assemblies. These advances have

Received: October 15, 2025. **Revised:** December 13, 2025. **Accepted:** January 22, 2026

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laid the groundwork for integrating protein structures into machine learning pipelines with unprecedented breadth and resolution.

Despite this progress, conventional modeling strategies still fall short in many real-world scenarios. Traditional structure-based approaches such as molecular docking and dynamics simulations often rely on rigid or pre-selected conformations, limiting their flexibility, scalability, and accuracy [7, 8]. On the other hand, sequence-based learning frameworks (e.g. compound–protein interaction prediction) lack spatial resolution and fail to capture the geometric and physicochemical complexity of binding interactions [9, 10]. Bridging this gap requires a new generation of structure-aware deep learning models capable of explicitly encoding and leveraging protein structural features across scales and modalities.

This review adopts a task-centric framework to trace the technological evolution of protein structure-aware deep learning in drug discovery [11]. We organize recent advances around three tightly coupled modeling goals:

Identifying where and how to bind: Covering binding site detection, structure-based screening, and pose prediction models that localize druggable regions and recover ligand conformations within protein pockets [12, 13].

Quantifying binding strength: Encompassing affinity prediction and interaction scoring models that learn spatial and energetic priors for assessing molecular complementarity.

Designing what to bind: Focusing on structure-informed molecular generation, where diverse generative frameworks, ranging from sequence-based to 3D coordinate-based models, construct novel ligands under protein constraints [14].

For each of these goals, we outline the key learning paradigms, structural representations, and modeling strategies, highlighting how geometric, topological, and physical features are progressively integrated. We further examine emerging co-folding frameworks that unify folding and binding in a single architecture, and discuss how they bridge the gap between sequence-based inference and structure-based learning. Finally, we present a comprehensive view of how protein structural priors are being embedded into generative models to drive target-specific molecular design, forming a continuum from interaction prediction to controllable 3D generation [15, 16]. In addition, we examine emerging co-folding frameworks that jointly model protein folding and ligand binding within a unified architecture, representing a shift from rigid docking to sequence-to-complex inference. These models offer enhanced flexibility and interpretability in scenarios where structural data are incomplete or absent.

Finally, we focus on the integration of protein structural constraints into molecular generation, where models aim to design novel ligands that are chemically valid, synthetically accessible, and structurally compatible with target binding sites. We present a unified taxonomy of structure-conditioned generation strategies—spanning sequence-based, graph-based, fragment-based, and full 3D coordinate-based models—and evaluate their advantages, limitations, and evolution.

Several excellent reviews have previously surveyed aspects of structure-based machine learning in drug discovery. Jiménez *et al.* [17] and Dhakal *et al.* [18] focused primarily on methodological advances in geometric deep learning and structure-based generative models up to early 2024, whereas more recent works such as Alakhdar *et al.* [19] and Bai *et al.* [20] have emphasized specific architectural paradigms or individual downstream tasks. In contrast, this review proposes a unified task-centric taxonomy that reframes structure-aware learning around five interdependent drug-discovery objectives, offering a

coherent roadmap for integrating protein structural information into both predictive and generative models. Figure 1 summarizes this task-centric framework, where multi-scale protein representations (atom, backbone, residue, and surface) underpin three goals: identifying where and how to bind [binding-site identification, pose prediction, and structure-based virtual screening (SBVS)], quantifying binding strength (interaction and affinity scoring), and designing what to bind (structure-conditioned molecular generation). By synthesizing methodological innovations across prediction and generation tasks, this review aims to guide future research toward models that are not only accurate and efficient, but also generalizable, interpretable, and experimentally actionable. This provides readers with a practical, decision-oriented framework for selecting and combining contemporary tools that is currently absent from the literature, thereby bridging the gap between rapidly evolving methods and their end-to-end application in real-world drug design.

To ensure that our survey captures the most relevant advances in the post-AlphaFold era, we applied clear criteria when selecting studies. The majority of included methods were published or released as widely adopted preprints between 2020 and 2025. As summarized in Table 1, these methods were required to incorporate explicit 3D structural information, address one of the core tasks defined in this review, provide reproducible methodological details, and report results on established community benchmarks.

Brief introduction of structure-informed machine learning

Protein structure as a foundation for drug discovery

Accurate protein structures form the molecular foundation for understanding drug–target interactions. From guiding the identification of druggable sites to enabling atomic-resolution modeling of binding poses, structural information plays a central role across nearly all stages of rational drug design.

Traditionally, structure-based drug discovery relied on experimentally determined protein conformations, typically through X-ray crystallography, nuclear magnetic resonance spectroscopy, or cryo-electron microscopy [26–29]. As structural data accumulated, computer-aided drug design emerged as a way to exploit available protein and ligand information in a more systematic and reproducible manner. Classical computer-aided drug design (CADD) produced several landmark therapeutics, including the rational design of saquinavir from HIV protease structures, the optimization of Factor Xa inhibitors such as rivaroxaban guided by pocket features, and the development of early hepatitis C virus protease inhibitors through transition state modeling of the NS3–NS4A complex. These successes demonstrated how structural insight, combined with computational modeling, could be translated into practical drug development strategies.

Despite these achievements, traditional CADD fundamentally depends on limited structural information, handcrafted molecular descriptors, and idealized assumptions about protein and ligand behavior. Ligand-based models struggle to encode three-dimensional electrostatics, conformational accessibility, and subtle electronic features, which restricts their ability to generalize to new scaffolds. Structure-based workflows rely heavily on a single high-quality protein conformation, making them sensitive to uncertainties in loop positioning, side-chain orientation, or induced fit motions. Classical

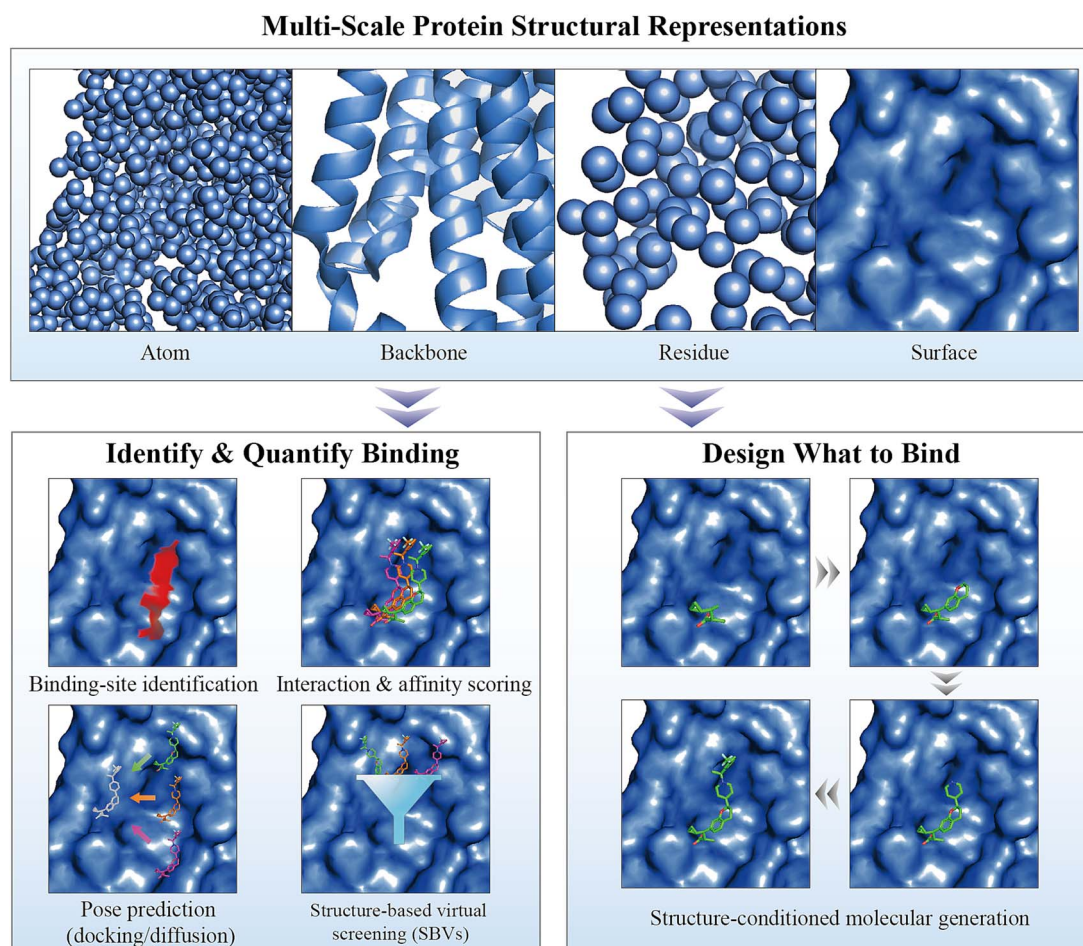


Figure 1 Structure-informed machine learning for drug discovery and design. (Top) Multi-scale protein structural representations, including atom, backbone, residue, and surface levels. (Bottom) Two core tasks of structure-informed machine learning for drug discovery and design: identify/quantify binding (left) and structure-constrained molecular generation (right).

Table 1 Criteria used to select papers included in this review.

Inclusion criteria

- Explicitly use 3D protein structural information (from experimental sources or predicted by AlphaFold2/3, ESMFold, RoseTTAFold, OmegaFold, etc.) as a core input or conditioning signal
- Directly address at least one of the five core tasks defined in this review (binding-site identification, pose prediction, interaction and affinity scoring, SBVs, and structure-conditioned molecular generation)
- Are accompanied by publicly available code or a detailed, reproducible methodological description
- Report performance on established community benchmarks (e.g. PDBbind, CASF, PoseBusters, CrossDocked, CASP/CAPRI ligand sets, etc.)

Exclusion criteria

- Purely sequence-based compound–protein interaction predictors without any 3D structural input
- Classical physics-based or empirical scoring/docking functions lacking a learned component
- Methods operating solely on ligand or small-molecule information without protein context
- Non-reproducible or purely retrospective benchmarking studies

scoring functions further approximate complex interaction physics, leading to unstable pose ranking and elevated false-positive rates, particularly for apo structures or flexible binding sites. Scalability is another constraint, as exhaustive docking, energy refinement, and molecular dynamics remain computationally expensive. Moreover, traditional CADD tools are generally static and do not easily incorporate new structural or biochemical information into iterative model updates.

Taken together, these limitations highlight the need for approaches that do not merely use protein structures as fixed inputs but are

capable of learning geometric, chemical, and dynamic regularities directly from large-scale structural data. This shift creates a natural opening for protein structure-aware machine learning, which treats structure as a learnable prior and applies it across prediction, scoring, and generation tasks with improved robustness and acceptability.

Breakthroughs such as AlphaFold2, RoseTTAFold, and ESMFold have made it possible to obtain high-quality protein structures computationally for entire proteomes [30–32]. Among them, AlphaFold2 represents a turning point, achieving near-experimental accuracy for a broad range of globular proteins. More recently, AlphaFold-Multimer

Table 2 Hierarchical representations and their task relevance.

Level	Key components	Representative features	Common use cases
Atom level	Heavy atoms (C, N, O, S, plus F/P/Cl/Br in special cases)	• Atomic properties: charge, hybridization, valence, electronegativity, vdW radius • Bond features: bond type, stereochemistry, H-bond donor/acceptor	Binding affinity prediction, pose evaluation
Backbone level	Peptide chain (N-C α -C=O)	• Dihedral angles (ϕ, ψ) • Secondary structure (helix, sheet, coil) • Connectivity, <i>cis/trans</i> conformation	Conformation modeling, fold classification
Residue level	20 natural amino acids	• AAindex-derived descriptors • Physicochemical descriptors (charge, hydrophobicity, volume, polarity), referring to dataset like AAindex [21] • Semantic embeddings from protein language models (e.g. ESM [22], TAPE, and Saprot [23])	Pocket prediction, interaction classification
Surface level	Solvent-accessible interface	• Geometric features: curvature, shape index, point normals [24] - Chemical features: electrostatics, hydrophobicity, charge distribution • Chemical features: charge distribution, hydrophilicity, and hydrophobicity [25]	Pocket detection, structure-conditioned generation

and AlphaFold3 extended this capability to protein complexes and small-molecule interactions, enabling modeling of full drug–target complexes *in silico* [6, 33]. These developments mark a fundamental shift: protein structures are no longer rare data assets but readily accessible resources that can be systematically integrated into AI-driven modeling pipelines.

This accessibility is particularly transformative for deep learning. In addition to directly enabling traditional virtual screening through docking or binding affinity estimation, protein structures can also serve as a source of diverse and informative features—such as residue-level contacts, binding surface topology, electrostatic fields, and conformational flexibility—that support a broader range of structure-aware learning tasks [34]. For instance, structure-guided pocket prediction models can identify druggable sites without requiring co-crystallized ligands, while structure-conditioned generative models are capable of designing novel compounds tailored to the geometry and physicochemical context of a given target [35, 36].

High-quality protein structure predictions have empowered AI drug discovery models by providing 3D structures as both inputs for more realistic interaction modeling and spatial priors encoding binding specificity and selectivity. This shift drives the need for effective structural representations that balance biological fidelity with computational feasibility—a focus we explore in the following section.

Featurization and modeling of protein structures

Multi-scale nature of protein structure

In this section, we hierarchically summarize protein structure featurization schemes (Fig. 2) to standardize their application, advantages, and complementarities in drug development. Protein structures are naturally hierarchical, with multiple levels of organization relevant to different modeling objectives. It should be noted that, in most resource-constrained or demand-limited situations, the full atomic representation of the complete protein structure may become computationally intractable. Consequently, researchers adopt coarse-grained or task-specific structural abstractions—ranging from atomistic detail to backbone traces, residue-level encodings, and surface geometries—to strike a balance between informativeness and efficiency (Table 2) [37].

Each level of representation captures distinct biochemical and spatial features while inevitably discarding certain types of information. For example, atom-level features precisely describe local chemical environments but typically focus only on partial protein regions (e.g. binding sites). Backbone or residue-level abstractions provide broader structural context but may blur fine-grained interaction patterns. Nonetheless, when applied judiciously, these multi-scale representations offer effective and interpretable structural priors for diverse drug design tasks.

These hierarchical representations each offer distinct advantages and limitations. Atom-level encodings provide fine-grained resolution required for modeling local complementarity and precise energetic interactions, but are often limited to small subregions due to computational cost. Backbone- and residue-level representations offer more scalable alternatives, enabling fold-level reasoning and generalized pattern recognition while sacrificing atomic detail. Surface-based representations strike a unique balance: they retain key geometric and chemical interface features relevant to ligand recognition, while reducing input complexity. However, surface abstraction may omit internal flexibility or allosteric context. Overall, selecting an appropriate representation hinges on the target task, available data, and computational budget. As we discuss next, these representations underpin three major structural modeling paradigms—graph-based, grid-based, and surface-based—that define how structural information is integrated into deep learning architectures.

Protein structure modeling

Once protein structures are appropriately represented, the next challenge lies in designing model architectures that can effectively process and learn from their spatial organization. The choice of modeling paradigm is tightly coupled with the input format: atom-level graphs, voxelized spatial grids, and molecular surfaces each afford distinct inductive biases and encode spatial relationships, geometric constraints, and structural features in different ways. Figure 2 illustrates these three key architectural paradigms—graph-based, grid-based, and surface-based—that collectively underpin modern protein structure-aware learning systems.

Graphs are a natural and efficient abstraction for protein structures, capturing spatial or chemical relationships in a flexible format. In

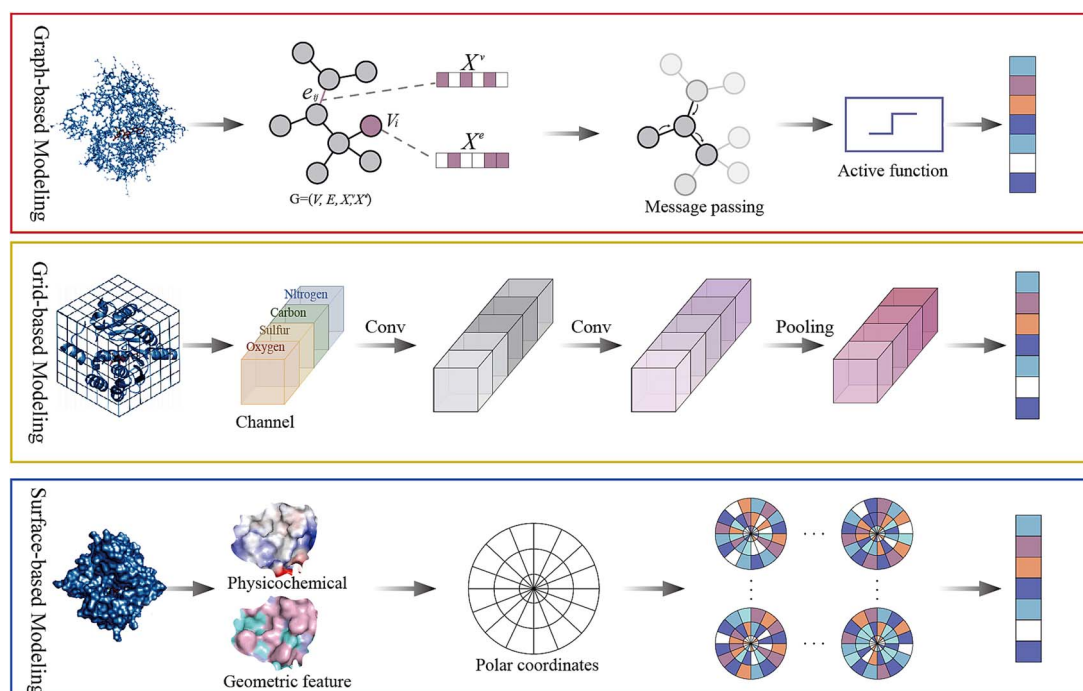


Figure 2 Architectures of protein structure-aware modeling. These models are based on graph-based (top), grid-based (middle), and surface-based (bottom) protein representation methods.

protein graphs, nodes typically represent atoms, residues, or backbone units, and edges encode pairwise proximity (e.g. distance cut-offs), physical bonds, or geometric features such as relative angles or orientations.

Graph neural networks (GNNs) have become the foundation for many structure-aware models. Classical message-passing frameworks—Graph Convolutional Network (GCN) [38], GraphSAGE [39], Graph Isomorphism Network (GIN) [40], and Graph Attention Network (GAT) [41]—enable contextual aggregation over neighboring nodes. These models capture local topological and spatial correlations, making them well suited for tasks such as secondary structure prediction, fold classification, or backbone reconstruction.

To better handle the 3D geometric nature of proteins, $SO(3)$ -equivariant GNNs [42] have been developed. These models preserve rotation and translation equivariance, ensuring that learned features are invariant to global rigid-body transformations. For instance, Geometric Vector Perceptrons (GVPs) operate on both scalar and vector channels in Cartesian coordinates, enabling vector-aware learning of orientation-sensitive features [43]; Geometric Bottleneck Perceptrons further integrate spatial priors to capture long-range, multi-body geometric dependencies [44]; Directed Weight Perceptrons introduce 2D/3D vector-based neurons and weight parameters to improve sensitivity to both global geometry and local frame relationships [45]. These architectures enable learning directly on the 3D molecular manifold and are particularly beneficial for capturing structural regularities such as symmetry, repeat motifs, and fold topology.

Grid-based approaches discretize the 3D space around a protein into voxels, where each voxel encodes local properties such as atom type, occupancy, charge, or hydrophobicity [46]. These representations align naturally with 3D convolutional neural networks (3D-CNNs), which apply spatial filters to learn hierarchical structural features [47, 48]. 3D-CNNs operate by stacking layers of convolution

and pooling operations, progressively capturing higher-level spatial patterns from local atomic neighborhoods. This enables direct modeling of volumetric shape, density distributions, and interaction cavities, making grid-based methods effective for tasks such as binding-site localization, structure similarity assessment, or local conformation analysis [49–52]. However, grid-based modeling suffers from key limitations: high memory cost and low scalability for full-protein or proteome-scale tasks, resolution trade-offs (finer grids improve accuracy but increase computation) and fixed frame assumptions that limit flexibility in modeling conformational ensembles [7]. To overcome these challenges, recent hybrid frameworks combine local voxel-based encodings with global graph reasoning, capturing both detailed spatial context and long-range dependencies while maintaining scalability [53].

Proteins exert their functions primarily through molecular interactions that occur at their surfaces. In structure-aware learning, accurate modeling of protein surfaces provides critical insights into binding-site geometry, interface shape complementarity, and electrostatic patterns, which are essential for downstream tasks such as pocket detection, interface matching, and ligand generation.

A representative framework, MaSIF, was introduced by Gainza *et al.* in 2019, where discretized protein surfaces are sampled into local patches and described using spatially aware geometric and chemical descriptors (e.g. shape index, curvature, electrostatic potential). These patches are processed via rotationally invariant geodesic convolutions to generate surface interaction fingerprints, enabling tasks like protein–protein interaction site prediction (MaSIF-site) and rapid surface-complementary docking (MaSIF-search) [24].

In 2021, Sverrisson *et al.* evolved this paradigm with dMaSIF, which replaces mesh discretization by constructing oriented point clouds directly from atomic coordinates [25]. By applying quasi-geodesic convolutions to these point clouds, dMaSIF captures surface features without explicit mesh generation, significantly improving

computational efficiency and robustness to structural noise (e.g. from predicted AlphaFold structures).

The latest advancement, MaSIF-neosurf, extends this surface-based framework to ligand-induced functional sites. It dynamically models the “neosurface” formed by protein–ligand complexes (e.g. Bcl2-venetoclax), integrating small-molecule chemical features into surface descriptors. This allows de novo design of ligands that reshape protein interfaces for novel molecular glues or allosteric modulators [54–56].

Surface-based models offer a unique balance between geometric richness and computational efficiency. Unlike voxel-based models, they operate directly on interaction-relevant features without modeling the internal atomic structure. Compared with graph-based models, they provide more accurate shape and interface context, especially useful in modeling binding selectivity or identifying cryptic pockets. However, they typically require additional pre-processing and are sensitive to structural noise, especially when applied to predicted (rather than experimental) protein structures.

Structure-informed drug-target binding modeling

Protein structures play a pivotal role in rational drug design, not only by revealing the spatial architecture of biological targets but also by enabling precise modeling of molecular recognition. Deep learning models that incorporate protein structural information have rapidly evolved to tackle key challenges in structure-based drug discovery. In this section, we adopt a task-centric view to summarize recent advances in structure-informed modeling. Figure 3 illustrates the overall framework, showing how different levels of protein structural representation provide inputs to four core tasks: binding-site identification (Section 0), binding strength modeling (Section 0), structure-based virtual screening (Section 0), and co-folding models (Section 0). To complement this conceptual view, Table 3 lists representative deep learning models under each task, together with their references.

Binding site identification

Identifying druggable binding sites on protein surfaces is a foundational step in structure-based drug discovery. These pockets represent accessible regions with suitable geometric and physicochemical environments for small-molecule binding. Classical computational tools have long addressed this problem using either geometric descriptors (e.g. surface curvature and cavity volume in Fpocket [57]) or energetic grids (e.g. probe–protein interaction energies in AutoSite [58]). However, such handcrafted heuristics are limited in generalizability and sensitivity to subtle structural variations.

A major shift occurred with the introduction of deep learning models operating on 3D voxelized representations of protein structures. These methods discretize protein space into regular grids and leverage 3D convolutional neural networks (3D-CNNs) to learn spatial patterns indicative of ligand-binding potential. DeepSite was among the first to demonstrate this approach, using voxel-wise classification to localize druggable pockets [59]. Later models built on this foundation: FRSite reformulated binding-site detection as an object detection task [60], Kalasanty introduced a 3D semantic segmentation framework [61], and DeepDrug3D incorporated interaction energy between protein residues and ligands into voxel features [62]. These models benefit

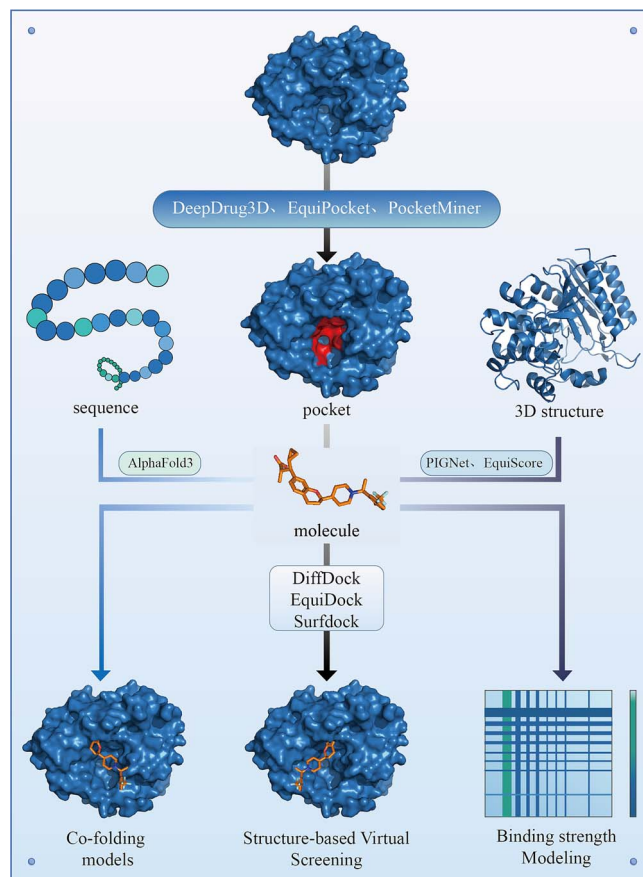


Figure 3 Overview of structure-informed drug-target binding modeling. The full workflow of structure-driven drug–target interaction analysis is presented in this figure, which starts with inputs (sequence, 3D structure, pocket) and employs tools (AlphaFold3, DiffDock) for co-folding, virtual screening, and binding strength modeling.

from compatibility with convolutional architectures and can effectively capture local physicochemical environments. However, they face scalability challenges due to high memory consumption, limited spatial resolution, and dependence on fixed coordinate frames.

To overcome these limitations, more recent approaches have turned to surface-based representations, which provide a compact and biologically meaningful description of protein interfaces. Models such as PInet treat the solvent-accessible surface as a point cloud, encoding each point with geometric and chemical features and processing them through hierarchical networks for local and global feature extraction [63]. DeepSurf adopts a similar strategy but applies *k*-means clustering to sample surface points and assigns each a set of 18 atom-derived descriptors [64]. PocketAnchor further enhances this representation by introducing anchor points on the surface and employing message passing to extract sub-pocket-level features [65]. These methods directly focus on the protein–ligand interaction interface and offer improved efficiency while retaining interpretability.

A parallel innovation has emerged through geometric deep learning frameworks that process raw 3D coordinates using rotation- and translation-equivariant architectures. EquiPocket pioneers the application of E(3)-equivariant GNNs to pocket prediction, operating on 3D point graphs without voxelization and exhibiting robustness to structural irregularities [66]. PocketMiner extends this idea to the

Table 3 Some protein structure-aware deep learning models in drug discovery.

Application	Model	Code	Level	Classes	Benchmarks
Binding-site identification	DeepDrug3D	https://github.com/pulimeng/DeepDrug3D	Atom	Grid	COACH420, HOLO4K
	DeepSite	https://www.playmolecule.org	Surface	Grid	COACH420, HOLO4K
	DeepSurf	https://github.com/stemylonas/DeepSurf.git	Surface	Surface	COACH420, HOLO4K
Binding strength modeling	EquiPocket	https://github.com/fengyuewuya/EquiPocket	Atom	SE(3)	COACH420, HOLO4K
	PINet	https://github.com/FTD007/PINet	Residue	Graph	COACH420, HOLO4K
	PocketAnchor	https://github.com/tiantz17/PocketAnchor/	Atom	SE(3)	COACH420, HOLO4K
	PocketMiner	https://github.com/Mickdub/gvp/tree/pocket_pred	Residue	Graph	COACH420, HOLO4K
	ScanNet	http://bioinfo3d.cs.tau.ac.il/ScanNet/	Residue + surface	Grid	COACH420, HOLO4K
	Fpocket	https://github.com/Weeks-UNC/fpocketR	Surface	Grid	COACH420, HOLO4K
	AutoSite	https://github.com/valueriver/AutoSite	Surface	Grid	COACH420, HOLO4K
	DrugBAN	https://github.com/peizhenbai/DrugBAN	Residue	Graph	PDBbind, CASF-2016
	DynamicBind	https://github.com/luwei0917/DynamicBind	Atom	SE(3)	PDBbind, PoseBusters
	EquiScore	https://github.com/CAODH/EquiScore	Atom	SE(3)	PDBbind, CASF-2016
	InteractionGraphNet	https://github.com/zjujdj/InteractionGraphNet/tree/master	Residue	Graph	PDBbind, CASF-2016
	NeuralPLexer	https://github.com/zrqiao/NeuralPLexer	Atom	SE(3)	PDBbind, PoseBusters
	MetalSiteHunter	https://github.com/ClinicalAI/metal-site-hunter	Atom	Graph	MetalPDB
	BindSiteS-CNN	https://github.com/Jing9558/BindSiteS-CNN	Atom	Grid	PDBbind
	PBCNet	https://github.com/BillCChen/PBCNet	Atom	Graph	PDBbind, CASF-2016
Structure-based virtual screening	PIGNet	https://github.com/ACE-KAIST/PIGNet	Residue	Graph	PDBbind, CASF-2016
	PSICHIC	https://github.com/huankoh/PSICHIC	Residue	Graph	PDBbind
	SgCPI	https://github.com/Y YingXia/SgCPI/	Residue	Graph	PDBbind
	STAMP-DPI	https://github.com/biomed-AI/STAMP-DPI	Atom	Graph	PDBbind
	TANKBind	https://github.com/luwei0917/TankBind	Atom	SE(3)	PDBbind, PoseBusters
Co-folding models	ZeroBind	https://github.com/myprecioussh/ZeroBind	Atom	SE(3)	PDBbind, CASF-2016
	DiffDock	https://github.com/gcorso/DiffDock	Atom	SE(3)	PoseBusters
	EquiDock	https://github.com/octavian-ganea/equidock_public	Backbone	SE(3)	PoseBusters
	SurfDock	https://github.com/RosettaCommons/rosetta	Surface	Surface	PoseBusters
	RosettaVS	https://github.com/Ro5-ai/hydrascreen	Atom	Physics	CASF-2016
	HydraScreen				
	RosettaFold	https://github.com/RosettaCommons/RoseTTAFold	Atom	SE(3)	PoseBusters, CASP15
	AphaFold3	https://github.com/google-deepmind/alphafold3	Atom	SE(3)	PoseBusters, CASP15
	Chai-I	https://github.com/chaidiscovery/chai-lab	Atom	SE(3)	PoseBusters, CASP15
	Boltz-2	https://github.com/jwohlwend/boltz	Atom	SE(3)	PoseBusters
	Protenix	https://github.com/bytedance/Protenix	Atom	SE(3)	PoseBusters
	NeuralPLexer3	https://github.com/iambic-therapeutics/	Atom	SE(3)	PoseBusters, CASP15
	RFdiffusion-AA	https://github.com/bakerla-boratory/rf_diffusion_all_atom	Atom	SE(3)	PoseBusters, CASP15

dynamic regime by training on molecular simulations that capture pocket-opening events. It uses GVP-enhanced GNNs to assign residue-level probabilities of cryptic site formation, enabling discovery of transient pockets that elude static structure-based methods [67]. Meanwhile, ScanNet offers a fully end-to-end framework that integrates atomic- and residue-level embeddings, local coordinate systems, and attention mechanisms. It learns hierarchical representations from raw atomic structures and provides interpretable predictions by attending to relevant spatial neighborhoods [68].

Together, these models illustrate a clear methodological evolution: from voxel-based spatial classification to surface-centric modeling, and finally to end-to-end geometric learning with built-in symmetry awareness and dynamic context. As accurate protein structures become increasingly available through tools like AlphaFold2/3, these structure-informed models promise scalable and precise identification of druggable pockets across the proteome, including shallow, flexible, or cryptic sites that were previously inaccessible to traditional methods.

Binding strength modeling

Predicting the binding strength (affinity) between small molecules and target proteins remains a cornerstone of rational drug discovery, with implications for virtual screening, lead optimization, and mechanistic understanding of molecular recognition. Recent advances in deep learning have increasingly incorporated structural and biophysical features to improve the fidelity and interpretability of affinity prediction models.

Early models primarily focused on graph-based representations of protein–ligand complexes, in which both molecular species are encoded as nodes and edges reflecting local structure or interaction context. InteractionGraphNet advances this paradigm by constructing three interrelated graphs—a ligand graph, a protein graph (restricted to binding-site atoms), and a bipartite protein–ligand graph—with atom-level attributes such as chemical type, spatial coordinates, and bonding patterns encoded to support dedicated intra- and inter-molecular convolutions [69]. In comparison, ZeroBind models protein structures at the residue level, where each amino acid node is endowed with spatial and physicochemical features and connected via proximity-informed edges. A novel Subgraph Information Bottleneck module is employed to identify minimal subgraphs representing potential binding pockets, thereby enhancing both computational efficiency and prediction accuracy [70].

To improve computational efficiency and data coverage, SgCPI adopts a localized sampling strategy. It extracts subgraphs centered around the predicted binding regions and encodes them using heterogeneous GNNs. This approach emphasizes the immediate protein–ligand interface while reducing the need for full-complex modeling, making it particularly suitable for large-scale screening scenarios [71]. Similarly, STAMP-DPI utilizes contact maps and TAPE-derived [72] residue embeddings to represent proteins as residue-level graphs and guide interaction scoring through semantic features [11].

While graph-based methods have demonstrated success in capturing geometric and topological relationships, subsequent models have introduced explicit physical and energetic priors to further improve generalization and interpretability. PIGNet incorporates parameterized energy functions—including van der Waals, hydrogen bonding, and electrostatic terms—into a molecular graph framework, enabling the network to learn physically meaningful representations of binding affinity. It further introduces an entropy-based loss component to enhance generalization across imbalanced datasets [73]. PBCNet proposes an alternating message-passing mechanism between pocket and ligand atoms and leverages graph attention incorporates physical and prior knowledge to prioritize key interaction pathways, particularly useful in ranking series of homologous ligands [74]. To capture more complex spatial interactions, TANKBind further introduces trigonometric constraints between protein structural blocks, integrating inter-block distance geometry to maintain spatial consistency and capture long-range dependencies within the protein–ligand complex [75]. Another representative approach, PSICHIC, models physicochemical constraints in interaction scoring from sequence-level inputs alone. By decoding interaction fingerprints through a specially designed GNN architecture, PSICHIC is capable of predicting binding affinity without relying on 3D structural data [76]. These designs mark a shift from purely data-driven architectures to physically grounded frameworks that align better with molecular interaction principles.

Recognizing the importance of spatial symmetry and conformational flexibility, recent efforts have introduced equivariant architectures capable of directly operating on 3D coordinates. NeuralPLexer combines multi-scale generative modeling with contact map prediction to reconstruct atomic-level representations of dynamic protein–ligand complexes. By avoiding voxelization and relying on SE(3)-equivariant operations, it maintains rotational and translational invariance while adapting to conformational diversity [77]. EquiScore combines heterogeneous GNNs with equivariant geometry to directly learn from raw 3D structures. It models local protein–ligand regions as heterogeneous graphs and incorporates protein–ligand interaction fingerprints as edge features. The network uses multi-head attention to highlight critical atom–atom pairs, enabling interpretable predictions of key interaction motifs that drive binding affinity [78]. Furthermore, DynamicBind formulates the binding process as a diffusion-driven traversal of the conformational energy landscape, producing pose and affinity estimates reflective of dynamic interaction trajectories [79].

A complementary direction aims to enhance model generalization under data scarcity and domain shift. DrugBAN integrates graph and convolutional encoders with bilinear attention and domain-adversarial training to learn transferable representations across diverse protein families [80]. EHIGN advances this approach through atomic-level interaction decomposition and explicit bias correction terms, effectively reducing the model's dependence on global graph pooling and improving performance in few-shot and cross-target prediction scenarios [81]. Building upon the established GNN framework discussed earlier, ZeroBind also incorporates a meta-learning paradigm that treats each protein target as a separate learning task. This architecture enables protein-specific model fine-tuning, significantly boosting predictive performance for novel or under-represented targets through task-adaptive parameter optimization [70].

Taken together, these developments illustrate a methodological evolution from rigid, graph-based encodings toward increasingly geometry-aware, physically grounded, and generalizable models. As structure-aware learning frameworks grow increasingly capable of modeling not only binding strength but also spatial orientation, they naturally converge with the goals of docking and pose prediction. In the next section, we shift from modeling binding interactions to their application in structure-based virtual screening, where pose accuracy, scoring reliability, and prioritization efficiency come together to accelerate drug discovery workflows.

Structure-based virtual screening

With increasing ability to capture protein–ligand interactions in both strength and geometry, deep learning models are now being integrated into SBVS pipelines. These workflows prioritize candidate molecules based on their predicted poses and affinities within target protein structures, enabling more targeted and scalable compound discovery.

Traditional SBVS frameworks typically consist of two components: pose generation—sampling plausible 3D conformations of ligands within protein binding pockets—and scoring functions to estimate binding affinity or interaction likelihood. Early computational tools relied on empirical or knowledge-based scoring functions combined with exhaustive or stochastic sampling strategies. For example,

RosettaVS integrates physical force fields with customized torsion potentials and atom types to evaluate interaction energies with high resolution [82]. More recent methods such as HydraScreen adopt an ensemble-based strategy, generating multiple ligand poses and ranking them using 3D convolutional neural networks trained on known complexes [83]. While these approaches highlight the value of incorporating structural cues into scoring, they often depend heavily on the quality and diversity of sampled poses, limiting robustness and efficiency.

To overcome the limitations of conventional rigid-body optimization and handcrafted scoring, deep learning models have recently shifted toward structure-aware pose prediction, wherein the protein–ligand binding process is learned directly from data. Among the earliest examples, EquiDock formulates pose prediction as a rigid-docking problem under geometric symmetry constraints. It represents proteins and ligands as graphs (typically at the residue or atom level) and applies SE(3)-equivariant neural networks to learn transformation-invariant features. During inference, EquiDock directly regresses the optimal rigid-body transformation matrix—predicting how a ligand should be rotated and translated to align with the target pocket. This approach bypasses traditional sampling and is especially efficient for blind docking tasks across large protein libraries [84].

Moving beyond rigid alignment, DiffDock treats pose prediction as a generative process over a structured pose manifold, encompassing ligand translation, rotation, and internal torsion angles. Instead of selecting from pre-sampled poses, DiffDock learns a diffusion-based model that incrementally constructs ligand conformations conditioned on the receptor. The model jointly represents ligand atoms, receptor residues, and receptor atoms as a heterogeneous geometric graph, enabling it to capture fine-grained interaction contexts. Importantly, it also includes a confidence model to rank sampled poses, providing both diversity and uncertainty quantification. The use of SE(3)-equivariant convolutions ensures robustness to coordinate choices and structural noise [85].

While both EquiDock and DiffDock operate under blind docking assumptions, newer models such as SurfDock incorporate explicit binding pocket information to improve pose localization and accuracy. SurfDock constructs a composite geometric graph comprising ligand atoms, protein surface points, and pocket residues, with node features encoding electrostatics, curvature, and language model–derived semantics. By leveraging a generative diffusion model, it optimizes ligand conformations not just in free space but within the spatial constraints of a predicted pocket, allowing pose sampling to be targeted and physically plausible. Crucially, SurfDock includes a pose scoring mechanism, enabling ranking and selection of predicted conformations based on learned interaction signals. This makes it especially useful for SBVS applications, where both accurate positioning and confident prioritization are essential [86].

Beyond improvements in average pose accuracy, reliable uncertainty quantification has become essential for deploying deep docking in prospective SBVS. In response, leading methods now incorporate explicit confidence modeling. DiffDock’s built-in confidence predictor regresses per-pose error and enables highly effective rank-by-confidence selection [85, 87], evidential deep learning heads provide calibrated predictive distributions in equivariant rigid-docking frameworks [84, 88], and ensemble-based approaches (deep ensembles or Monte-Carlo dropout) are used in NeuralPlexer, DynamicBind, and SurfDock [77, 79, 86]. Calibration is generally satisfactory on high-confidence AlphaFold regions (median pocket

pLDDT >90), but degrades sharply in lower-confidence or flexible areas.

Practical guidelines for trusting blind docking outputs on predicted structures are now emerging from large-scale benchmarks [36, 89, 90]:

- Trustworthy (>70% Top-1 RMSD <2 Å): rigid orthosteric pockets in globular domains with median pLDDT >90 and predicted aligned error (PAE) <5 Å (e.g. kinase ATP sites, GPCR transmembrane bundles).
- Use with caution (40%–60% success): pockets with pLDDT 70–90, shallow interfaces, or minor induced fit.
- Unreliable and prefer pocket-conditioned fallback (<30% success): low-pLDDT loops (<70), cryptic pockets requiring opening, protein–protein or allosteric interfaces, macrocycles, or covalent warheads.

These advances in pose generation, confidence-aware ranking, and hybrid blind/pocket-conditioned strategies have directly expanded the scope and reliability of modern SBVS pipelines. Deep learning frameworks now support truly end-to-end workflows that integrate structure-aware pose generation, calibrated uncertainty estimation, and large-scale virtual screening—transforming pose prediction from an isolated pre-processing step into the central engine of structure-based drug discovery.

Co-folding models

The traditional pipeline for modeling protein–ligand complexes often decouples structure prediction and molecular docking, relying heavily on experimentally determined protein structures or extensive sampling to estimate binding poses. Recent advances in deep learning have enabled a new class of implicit structure-aware models that predict protein–ligand complex structures without relying on experimentally resolved inputs. By jointly modeling folding and binding, these models unify what were traditionally treated as separate stages—protein structure prediction and ligand docking—into a single co-folding framework, offering a more data-efficient and biologically realistic approach.

Compared to conventional docking pipelines, co-folding models offer distinct advantages: they implicitly accommodate induced fit, cryptic pocket formation, and binding-coupled folding, all without the need for exhaustive conformational sampling. This enables applications such as mutation effect modeling, cryptic site targeting, and lead prioritization under limited structural information—scenarios where traditional pipelines often struggle.

These conceptual advances have spurred the development of several all-atom co-folding frameworks, including RosettaFold All-Atom [32], AlphaFold3 [6], Chai-1 [91], Boltz-1 [92], Protenix [93], NeuralPlexer3 [71], and Boltz-2 [94], for example. These models are poised to play a central role in next-generation structure-based virtual screening and AI-driven drug design.

Among representative models, Uni-Mol [95] learns 3D-aware embeddings of proteins and ligands via Transformer-based architectures, using pairwise distance maps and atom-level features. It bypasses the need for full atomic structures while achieving strong performance on tasks such as affinity prediction and pose estimation, particularly in settings with incomplete or predicted structural data.

AlphaFold3 is a landmark in this domain. Extending AlphaFold2’s capabilities, it models multimolecular assemblies—including proteins, RNAs, DNAs, and small molecules—within a unified framework. By leveraging co-evolutionary signals and spatial constraints, AlphaFold3

can predict accurate binding poses from sequence alone, supporting sequence-to-complex modeling that broadens the reach of AI-driven drug design.

Boltz-2 represents a recent leap forward in this category. In addition to accurate complex structure prediction, it integrates an explicit affinity prediction module trained to match the performance of physics-based methods such as free energy perturbation. Boltz-2 also supports user-defined contact constraints and template conditioning, enabling flexible and interpretable modeling of multi-chain and multi-ligand complexes—capabilities crucial for rational design and hypothesis testing.

Despite the impressive headline numbers on PDBbind-derived or CASP15/CAPRI-like test sets, co-folding models exhibit several systematic failure modes that are only now being rigorously quantified. First, performance drops dramatically for out-of-distribution (OOD) ligands—especially macrocycles, organometallics, covalently bound warheads, and ligands with >50 heavy atoms—which are severely under-represented in the PDB training corpus, raising concerns over data leakage and overfitting [96]. Second, severe overfitting and data leakage remain pervasive: many benchmarks contain near-duplicate complexes, homologous receptors that share >30% sequence identity with training entries, or even exact protein–ligand pairs that appear in both training and test sets after naïve random splitting. Such leakage artificially inflates pose RMSD < 2 Å success rates by 20%–60% and binding-affinity Pearson *R* by 0.15–0.30 in recent analyses. Third, because most models are trained predominantly on high-resolution crystal structures (median ~2.0 Å), they often fail on lower-confidence regions of AlphaFold-predicted pockets or on induced-fit/cryptic-pocket cases that deviate from the apo template [89, 97, 98].

To mitigate these issues and obtain trustworthy benchmarks, the community is converging on stricter evaluation protocols: (i) receptor-level or fold-level clustering (e.g. sequence identity <30% between train/val/test receptors), (ii) ligand-scaffold deduplication using Tanimoto thresholds on ECFP4 fingerprints, (iii) time-split datasets that exclude any complex solved after the model's training cutoff, and (iv) dedicated OOD test sets such as PoseBusters-Outliers and Molecule-ULTIMATE. Adopting these practices is essential before co-folding models can be confidently deployed in prospective drug-discovery campaigns targeting novel or difficult chemical series [99].

Protein structure-informed molecular generation

As protein structural data becomes increasingly accessible and accurate, the incorporation of such information into molecular generative models has transformed the landscape of AI-driven drug design. Modern approaches (Table 3) condition the generative process on the structural features of protein binding sites—ranging from coarse cavity shapes to fine-grained atomic geometries—thus enabling the synthesis of ligands that are not only chemically valid but also spatially and functionally compatible with their targets. This structure-aware paradigm shifts molecular design from target-agnostic exploration toward precise, mechanism-driven generation.

Figure 4 intuitively summarizes this structural conditional generation paradigm, in which protein binding site features serve as core constraints, guiding four key molecular generation strategies: sequence-based 3D conditional generation, fragment-based ligation and growth, graph-based generation under structural constraints, and 3D coordinate-based generation (including diffusion models). These categories represent a progressive evolution in how structural

constraints are modeled and utilized—from latent conditioning on pocket embeddings, to discrete graph construction guided by pocket geometry, to full 3D conformation generation within protein-shaped spatial priors. Each strategy offers distinct trade-offs in terms of efficiency, flexibility, and spatial fidelity, collectively forming a versatile toolkit for rational drug discovery.

Sequence-based generation with 3D conditioning

Sequence-based generative models typically construct molecules as SMILES or SELFIES strings using molecular language models. Historically, these methods operated independently of structural constraints, limiting their ability to produce ligands that are geometrically compatible with specific protein targets. Recent advances have addressed this gap by incorporating structural information—particularly protein binding pocket features—into the generative process, bridging 1D sequence outputs with 3D spatial requirements.

Early work by Xu *et al.* introduced a variational autoencoder that conditioned SMILES generation on pocket-level descriptors such as Coulomb matrix eigenvalues and amino acid composition [100]. While the molecular output remained two-dimensional, the incorporation of 3D-derived features into the latent space allowed for implicit structural bias toward pocket-compatible chemotypes.

Subsequent models improved spatial fidelity and learned direct protein–ligand geometric relationships. 3DSMILES-GPT [101] encoded atomic positions and pocket features using Gaussian distance matrices, integrating these into a Transformer decoder that jointly attends to token-level chemical features and 3D spatial proximity. Trained with reinforcement learning on property-annotated molecules, the model enhanced drug-likeness, synthesizability, and binding relevance.

TamGen [102] similarly employed a GPT-like architecture, incorporating a protein encoder and a distance-aware attention mechanism. This design allowed for the direct modeling of spatial interactions between the ligand SMILES tokens and the protein pocket, supporting activity-specific and structure-aware generation.

Lingo3DMol [103] pushed the boundary further by introducing FSMILES, a hybrid representation combining molecular fragments with spatial tokens. This model was pretrained on fragment-pocket binding patterns and enabled the generation of 3D-conformationally plausible SMILES sequences with spatial awareness.

BindGPT [104] unified multiple generation tasks—3D molecule design, conformation prediction, and protein-conditional generation—within a large language model architecture. By jointly encoding SMILES and atomic coordinates, and leveraging pretraining on large-scale datasets with reinforcement tuning, BindGPT achieved performance approaching 3D diffusion models with significantly reduced computational cost.

Sequence-based models have progressed from simple latent conditioning (e.g. Xu *et al.*) to fully attention-driven spatial integration (e.g. TamGen, 3DSMILES-GPT) [101, 102], and more recently to hybrid representations that incorporate explicit spatial tokens (e.g. Lingo3DMol, BindGPT) [103]. This reflects a growing convergence between molecular language models and structure-aware design, leveraging pretrained language representations to address the demands of structure-based drug discovery.

Across these models, the common thread is the strategic use of lightweight 3D conditioning—whether through distance matrices, geometric encoders, or fragment-level embeddings—on top of scalable, interpretable sequence generation frameworks. While

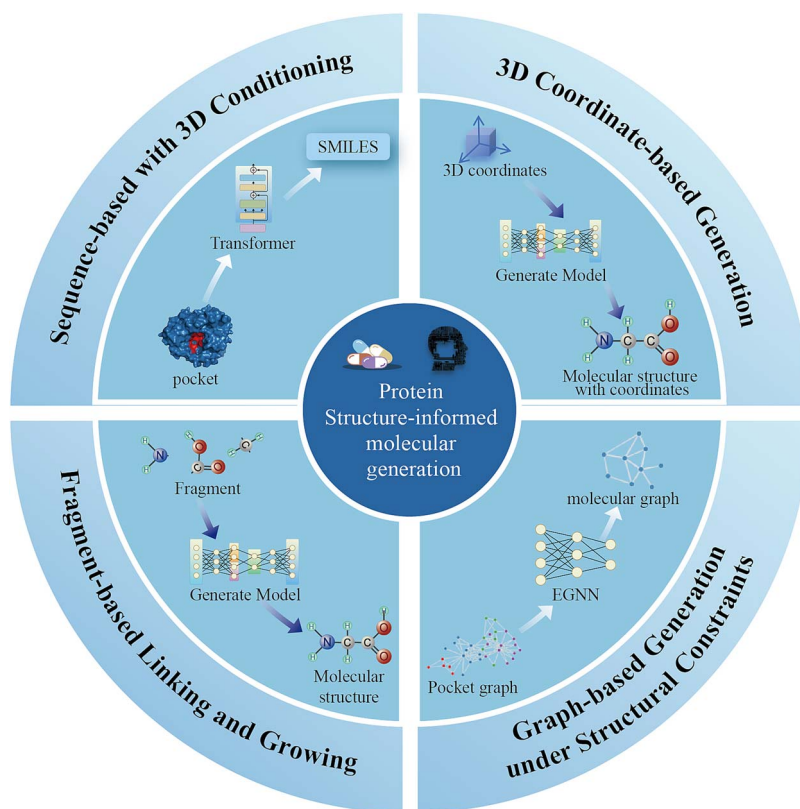


Figure 4 Strategies for protein structure-informed molecular generation. Four core strategies for protein structure-informed molecular generation are illustrated, namely sequence-based generation with 3D conditioning, 3D coordinate-based generation, graph-based generation under structural constraints, and fragment-based linking and growing, along with their corresponding key tools and workflows.

the outputs remain in 1D chemical space, these models deliver several advantages: fast inference, efficient training, compatibility with reinforcement learning, and transparent mapping between generated substructures and structural drivers. While less precise than coordinate-level generation, these models are highly suitable for early-stage virtual screening and scaffold exploration. Their speed, flexibility, and interpretability make them valuable upstream tools in hierarchical pipelines, where they provide chemically meaningful seeds for subsequent 3D refinement and docking.

Fragment-based linking and growing

Fragment-based molecular generation builds drug-like compounds by assembling or extending chemically meaningful substructures under the structural constraints of a protein target. Rooted in medicinal chemistry practices, this strategy balances design flexibility and chemical realism, offering modular control over scaffold topology and binding specificity. Early methods primarily operated in 2D space—generating molecular graphs or SMILES strings—but increasingly, models have adopted full 3D representations, guided by the geometry of protein pockets.

Initial efforts such as SyntaLinker [105] employed SMILES-based Transformers conditioned on pharmacophoric constraints like shortest linker bond distance or functional group placement. Though operating in 1D chemical space, these models allowed indirect incorporation of protein pocket information via user-defined rules. Subsequent models like DeLinker [106] introduced more explicit spatial awareness, using geometric features—such as inter-fragment distances and

angles—to guide 2D linker generation within protein-defined anchor constraints. DEVELOP [107] extended this framework by encoding the pocket with 3D CNNs and incorporating pharmacophoric grids, improving geometric fidelity and controllability.

A significant evolution occurred with the emergence of models generating full 3D molecular structures. 3D-Scaffold demonstrated how predefined scaffolds, represented by their atomic coordinates, could be elaborated into synthetically feasible candidates while preserving their orientation in space. DeepLigBuilder [108] further refined this paradigm by growing entire molecules atom-by-atom from a core, directly using protein-derived spatial constraints. These approaches introduced atomic resolution into fragment elaboration, enabling better fit within target pockets.

As models progressed, they began to blend generation with optimization. FRAME applied reinforcement learning to adaptively expand ligands, learning from pocket–ligand interactions which fragment to add and where to attach it. DiffLinker [109] employed E(3)-equivariant diffusion models to produce 3D-compatible linkers with high structural fidelity. FragGen [110] pushed the frontier further by introducing a deep geometric framework capable of end-to-end 3D fragment linking optimized for spatial fit, validated through applications in kinase inhibitor design.

In parallel, hybrid models emerged. AutoFragDiff [111] combined fragment-wise diffusion and autoregressive generation to construct ligands around a given scaffold and pocket geometry, guided by anchor atoms. Zhou *et al.* proposed a modular decomposition approach that splits ligands into arms and cores, then refines their reassembly in a spatially aware loop. These models not only generate

chemically valid molecules but also optimize fragment orientation and interactivity through learned 3D refinement.

The fragment-based paradigm has also expanded to address multi-target drug design. AlxFuse [99] exemplifies this direction by encoding multiple protein–ligand complexes, extracting interaction-informed pharmacophore trees, and guiding compound assembly through Monte Carlo Tree Search. It enables the generation of ligands capable of dual engagement—an emerging goal in polypharmacology and systems therapeutics.

Overall, fragment-based models have evolved from heuristic 2D assemblers to highly expressive 3D generators tightly integrated with protein structure. Their progression reflects a growing emphasis on spatial precision, chemical validity, and optimization capabilities. Compared to SMILES-based models, they offer greater interpretability and scaffold control; compared to end-to-end 3D coordinate generators, they are more modular and sample-efficient. These strengths make fragment-based generation especially suitable for scaffold hopping, lead optimization, and constrained analog design in the intermediate stages of drug discovery.

Graph-based generation under structural constraints

Graph-based molecular generation leverages the structural expressiveness of molecular graphs while incorporating explicit constraints from protein target geometry. These models construct molecules as atom–bond graphs and condition the generation process on 3D information derived from protein binding sites, typically represented as atomistic graphs or point clouds. As a result, they are capable of generating chemically valid, spatially compatible molecules tailored to complex topological and steric environments.

Early works like Pocket2Mol [112] and PocketFlow [113] exemplify this paradigm. Both models adopt E(3)-equivariant GNNs to encode the full 3D geometry and chemical features of the protein pocket. In Pocket2Mol, the pocket is encoded as an atom-level spatial graph, which then conditions an autoregressive molecular graph decoder that predicts atom and bond types in a masked generation setup. PocketFlow refines this by introducing joint pocket–ligand representations and incorporating domain knowledge into both encoding and decoding, improving both the fidelity and chemical relevance of generated molecules. These models generate 2D molecular graphs, but their use of 3D pocket constraints ensures spatial complementarity in the final structure.

Further developments enhanced the geometric accuracy and stepwise control of molecular assembly. GraphBP [114] constructs molecules atom-by-atom using local coordinate frames, allowing for the fine-grained placement of each atom relative to the protein surface. This enables the model to generate 3D-consistent 2D molecular graphs that align with the geometry of the pocket. DrugHIVE [115] takes a hierarchical approach using variational autoencoders, learning coarse-to-fine interaction patterns between ligands and receptors and enabling the generation of chemically diverse and receptor-specific molecules.

Meanwhile, models like cG-SchNet [116] extend autoregressive generation to the 3D domain by predicting interatomic distances in tandem with graph connectivity. The resulting graphs are thus inherently 3D-aware, even if not explicitly generating Cartesian coordinates. Atz *et al.* take this further by integrating structural interactome graphs, combining graph Transformers with long short-term memory modules. They convert the protein pocket into a 3D graph and

condition molecular generation on both sequence-derived and geometric binding information. This model exemplifies the move toward hybrid modalities that jointly learn from topology, sequence, and spatial constraints.

Graph-based generation has progressed from basic pocket-conditioned graph decoders to sophisticated equivariant GNNs with autoregressive sampling and hybrid multi-modal integration. Across these models, the common theme is the use of 3D pocket structure not just as context, but as an active constraint guiding atom placement, bond formation, and scaffold growth. While their outputs are typically 2D molecular graphs, their training objectives and architectural designs incorporate 3D-aware loss terms, such as geometric distances or local frame consistency, ensuring that generated molecules are not only chemically valid but spatially plausible.

Compared to sequence-based models, they provide better control over molecular topology; compared to 3D coordinate generators, they are often more sample-efficient and easier to train. As such, graph-based generation under structural constraints serves as a versatile middle ground—balancing spatial fidelity, molecular realism, and computational tractability—making it a critical component of the broader landscape of structure-informed molecular design.

3D Coordinate-based generation

Unlike graph- or sequence-based approaches that typically operate in 2D chemical space, 3D coordinate-based generation directly models the Cartesian coordinates of atoms to construct fully resolved molecular conformations. These models aim to generate molecules that not only satisfy chemical validity but also physically align with the spatial environment of a given protein binding site. They vary significantly in architecture—from geometric deep learning over voxelized grids or point clouds to generative processes such as energy minimization or denoising diffusion—and increasingly serve as the backbone of structure-based *de novo* design. Table 4 summarizes representative protein structure-aware deep learning models for molecular generation, covering sequence-based, fragment-based, graph-based, and 3D coordinate-based approaches.

Early non-diffusion models explored diverse strategies for modeling 3D molecular geometry in a protein-constrained context. Ragoza *et al.* proposed a CVAE framework [117] that voxelizes protein–ligand complexes into atomic density grids, generating ligand grids conditioned on receptor embeddings, and later reconstructing molecules via atom fitting and bond inference. DEVELOP further advanced this idea by encoding pharmacophoric and spatial features of the binding site into a voxelized density grid, using it to guide atomic coordinate prediction within a latent generative network. Such approaches emphasized holistic 3D shape complementarity but faced challenges in converting dense grids into discrete molecular graphs.

In parallel, point cloud-based models like PMDM [118] reframed both protein pockets and ligands as interacting sets of 3D points, connected by virtual edges simulating physical forces (e.g. van der Waals, electrostatics). Rather than relying on sampling from latent space, PMDM employs iterative spatial transformations to progressively assemble molecular structures that align with the pocket's local environment. SurfGen [119] pursued a surface-aware strategy by treating the protein binding site as a triangulated mesh and optimizing atom placement along geodesic distances and electrostatic fields, thus sculpting ligands directly into the 3D mold of the target.

Table 4 Some protein structure-aware deep learning models in molecular generation.

Application	Model	Code
Sequence-based generation with 3D conditioning	3DSMILES-GPT	https://github.com/ashipiling/GPT_3DSMILES
	TamGen	https://github.com/SigmaGenX/TamGen
Fragment-based linking and growing	Lingo3DMol	https://github.com/stonewiseAIDrugDesign/Lingo3DMol
	BindGPT	https://github.com/bindgpt/bindgpt.github.io
	ConfGF	https://github.com/DeepGraphLearning/ConfGF
	SyntaLinker	https://github.com/YuYaoYang2333/SyntaLinker
	DeLinker	https://github.com/fimrie/DeLinker
	DeepLigBuilder	https://disk.pku.edu.cn:443/link/
	DiffLinker	https://github.com/igashov/DiffLinker
	FragGen	N/A
	AutoFragDiff	https://github.com/keiserlab/autofragdiff
	Pocket2Mol	https://github.com/pengxingang/Pocket2Mol
Graph-based generation under structural constraints	PocketFlow	https://github.com/Saoge123/PocketFlow
	GraphBP	https://github.com/divelab/GraphBP
	DrugHIVE	https://github.com/jssweller/DrugHIVE
3D Coordinate-based generation	cG-SchNet	https://github.com/atomistic-machine-learning/cG-SchNet
	CVAE	https://github.com/timbmg/VAE-CVAE-MNIST
	PMDM	https://github.com/LayneHuang/PMDM/tree/main
	SurfGen	https://github.com/HaotianZhangAI4Science/SurfGen
	TopMT-GAN	https://github.com/aeghnsw/TopMT
	VoxBind	https://github.com/Genentech/Voxbind
	DiffBP	https://github.com/opipari/diffBP
	BindDM	https://github.com/YangLing0818/BindDM
	IPDiff	https://github.com/YangLing0818/IPDiff
	DiffSBDD	https://github.com/arneschneuing/DiffSBDD
	DiffInt	https://github.com/sekijima-lab/DiffInt
	FlexSBDD	https://github.com/zaixizhang/FlexSBDD
	DynamicFlow	https://github.com/andreacasalino/DynamicFlow
	Apo2Mol	N/A

Topological approaches using adversarial frameworks also emerged. TopMT-GAN [120] encodes pocket topology using layered spheres and learns 3D molecular topologies through graph-to-graph translation and adversarial type assignment. VoxBind [121] leverages 3D voxel-denoising networks under a neural empirical Bayes framework to iteratively recover ligands from random initializations. These models balance fine-grained 3D control with sampling flexibility, making them well suited for topology-driven design.

The advent of diffusion-based generation marked a major leap in 3D molecular modeling. These models treat generation as a reverse-time denoising process that reconstructs atomic coordinates and identities from Gaussian noise, while conditioning on structural features of the target protein. DiffBP [122] pioneered this direction by enabling all-atom ligand generation within a pocket point cloud. Later methods such as BINDDM [123] integrated SE(3)-equivariant neural networks and binding subcomplex graphs to better capture the hierarchical spatial relationships between receptor and ligand.

IPDIFF introduced a novel use of interaction priors—learned from protein–ligand complexes—to dynamically modulate the denoising trajectory, yielding improved geometric realism and binding affinity. DiffSBDD extended this framework with stronger SE(3)-equivariant architectures, while PILOT integrated multi-objective sampling strategies, such as synthetic accessibility and docking scores, to bias generation toward experimentally relevant chemical space. DiffInt [124] went a step further by modeling hydrogen bonds as explicit interaction particles within the diffusion field, enhancing its ability to reconstruct meaningful non-covalent networks.

Collectively, these models mark a shift from indirect shape matching toward 3D coordinate-based generative frameworks that more

faithfully capture ligand flexibility and pocket-induced fit. In practice, such coordinate-level generators are best used as high-precision components for pose refinement and structure-guided optimization, complementing faster graph or sequence-based models upstream.

Despite the progress enabled by recent coordinate-based generative models, most SBDD pipelines still treat the protein binding site as static and rigid [125, 126]. This assumption ignores the intrinsic flexibility of molecular recognition, confines design to a single holo geometry, and can miss ligands that stabilize alternative, and sometimes more favorable, pocket conformations [127]. The challenge becomes even more acute for targets with only apo structures available, where the lack of a suitable bound-state template often leads to poor ligand quality.

To overcome these limitations, a new wave of models explicitly couples ligand generation with pocket flexibility. DynamicFlow [128] introduces a full-atom stochastic flow-matching framework trained on MISATO molecular-dynamics trajectories [129], enabling simultaneous prediction of holo-like pocket rearrangements and the bound ligand starting from an apo structure. FlexSBDD [130] adopts a related flow-matching strategy using an E(3)-equivariant architecture with scalar–vector features, supplemented by structure relaxation and side-chain repacking to expose the model to realistic backbone and side-chain movements. Apo2Mol [131] extends this direction with a hierarchical all-atom diffusion model that jointly samples ligand geometry and pocket conformational changes via a coupled flow. Trained on >50 000 apo–holo pairs from PDB and AlphaFold3, it learns residue-level plasticity—including torsional variability via Gaussian-mixture priors—and mitigates the size mismatch and affinity instability that often limit rigid-pocket SBDD. Beyond

ligand-centric approaches, MaSIF-neosurf [54] extends flexibility-aware modeling to the protein–ligand interface by learning “neosurfaces” induced upon binding, enabling the design of complementary protein or peptide binders shaped by these dynamic interaction landscapes.

Taken together, these developments show that 3D coordinate-based generation has matured from early geometric heuristics into a unified class of physically grounded generative frameworks capable of capturing molecular flexibility, pocket-induced fit, and spatial constraints with far greater fidelity than graph- or sequence-based approaches. As such, it now provides the structural backbone on which increasingly realistic protein-aware design workflows are built.

Accuracy and acceptability: canonical structure prediction versus protein-aware generative design

Traditional canonical structure-based drug design—typically involving rigid or semi-flexible docking (e.g. AutoDock Vina or Glide) on static crystallographic or AlphaFold-predicted templates—excels in speed and interpretability for well-defined orthosteric sites but falters on dynamic targets. On the PoseBusters-AF2 benchmark [89], such methods achieve ~55% Top-1 pose accuracy (RMSD <2 Å) for rigid pockets but drop to <25% for induced-fit or cryptic sites, where single-conformation templates fail to capture loop flexibility or multi-body rearrangements. Affinity predictions further suffer, with Pearson *R* correlations ~0.4–0.6 against experimental ΔG on PDBbind-v2020, often overestimating stability due to neglected entropic penalties [132].

In contrast, protein-structure-aware generative models (e.g. diffusion-based 3D frameworks like DiffSBDD, Apo2Mol, or co-folding-integrated generators like Boltz-2) treat structures as learnable priors, yielding superior geometric plausibility and multi-modal sampling. These approaches boost pose success to 70%–85% on flexible benchmarks (e.g. CrossDocked2020 flexible subset), better modeling induced fit via equivariant flows or joint protein–ligand optimization [133]. Regarding acceptability, generated ligands exhibit higher synthetic feasibility (SA scores >0.7 versus 0.5 for docking outputs) and downstream compatibility (e.g. 20%–30% fewer MD instability flags), though they require additional filtering for novelty and ADMET compliance.

Overall, while canonical methods remain the gold standard for high-throughput validation, generative designs offer greater accuracy for challenging targets at the cost of ~10–100× higher inference time—hybrid pipelines combining both are emerging as the optimal path forward.

Challenges and outlook

Despite the remarkable progress in protein structure-aware molecular design, several key challenges remain before these models can be reliably deployed in real-world drug discovery pipelines. A primary limitation lies in data availability and distributional bias. While structure prediction models have made vast protein conformational landscapes accessible, high-quality protein–ligand complex data—especially for underexplored targets or non-canonical binding modes—remain sparse. This restricts model generalization and increases the risk of overfitting, particularly for co-folding and 3D generative frameworks.

Another critical bottleneck is the lack of rigorous benchmarking protocols. Current evaluation often relies on retrospective datasets or

proxy metrics (e.g. docking scores, ligand RMSD), which do not always correlate with experimental activity. There is a pressing need for standardized, prospective benchmarks that account for ligand novelty, binding mode plausibility, and synthetic feasibility in a structure-aware setting. Moreover, cross-task benchmarking—e.g. evaluating generative models not only on geometry but also on their impact on downstream tasks such as affinity ranking or mutational robustness—remains underdeveloped.

Model design also faces a trade-off between interpretability and expressiveness. While physics-inspired and equivariant architectures improve geometric fidelity, they often do so at the expense of transparency and controllability. Bridging symbolic chemical knowledge (e.g. pharmacophores, reaction rules) with neural architectures remains an open direction toward human-in-the-loop design and hypothesis generation.

Looking forward, the field is moving toward integrated, multi-objective frameworks that unify structure prediction, binding modeling, and molecule generation into coherent pipelines. Emerging trends include foundation models trained on multi-modal biochemical data, reinforcement learning loops with structural feedback, and simulation-augmented generative models that blend machine learning with physical realism. Furthermore, flexibility modeling—at both protein and ligand levels—will be critical for targeting dynamic or cryptic pockets and for predicting resistance-associated conformational shifts.

Ultimately, protein-aware molecular generation represents a paradigm shift—from structure as a static input to structure as a dynamic, learnable prior. As methods continue to evolve from heuristic rules to data-driven physical reasoning, they promise to reshape the landscape of rational drug design—making it faster, more informed, and increasingly programmable.

Key Points

- This review adopts a task-centric perspective to synthesize structure-informed machine learning for drug discovery, organizing advances around three core goals: identifying binding sites/poses, quantifying binding strength, and designing target-compatible ligands, all supported by multi-scale protein representations (atom, backbone, residue, surface).
- Breakthroughs in protein structure prediction (e.g. AlphaFold2/3, ESMFold) have transformed structural data from rare experimental assets into accessible resources, enabling the shift from traditional rigid docking/dynamics to geometry-aware deep learning models (e.g. SE(3)-equivariant GNNs, diffusion models) that capture spatial and physicochemical complexity.
- Key advances in binding modeling include (i) binding-site identification (e.g. EquiPocket, PocketMiner) leveraging surface-based or geometric deep learning; (ii) affinity prediction integrating physical priors and equivariant architectures (e.g. PIGNet, EquiScore); (iii) virtual screening with pose generation/scoring (e.g. DiffDock, SurfDock); and (iv) co-folding models (e.g. AlphaFold3, Boltz-2) that unify protein folding and ligand binding.
- Four major structure-conditioned molecular generation strategies are classified: sequence-based 3D conditioning, fragment-based linking/growing, graph-based generation under structural constraints, and 3D coordinate-based generation (notably diffusion

models), each balancing chemical validity, spatial fidelity, and computational tractability.

- Critical challenges remain (data scarcity, biased benchmarks, interpretability–expressiveness trade-offs), with future directions focusing on multi-objective integrated pipelines, flexibility modeling (for dynamic/cryptic pockets), and merging machine learning with physical realism to enhance drug design practicality.

Author contributions

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Conflict of interest

The authors declare no conflicts of interest.

Funding

This work was supported by the National Natural Science Foundation of China (62366002, 82204278, T2225002, and 82273855), the National Key Research and Development Program of China (2023YFC2305904 and 2022YFC3400504), the Strategic Priority Research Program of the Chinese Academy of sciences (XDB0830000), the Key Technologies R&D Program of Guangdong Province (2023B1111030004), and the Lingang Laboratory (LGL-8888).

Data availability

Data sharing is not applicable to this article as no new datasets were created or analyzed during the current study.

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